JESSICA WAN

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EDUCATION:

Brown University, *Providence, RI* | *Master of Science, Computer Science* (Fall 2024 - May 2025)
Brown University, *Providence, RI* | *Bachelor of Science, Computer Science* (Fall 2020 - May 2024)

WORK EXPERIENCE:

- **C++ Software Engineer,** *Dassault Systemes – Johnston, RI* (June 2025 - Present)
  - Built automated unit and integration tests in C++/Shell, boosting coverage by > 50% and ensuring application stability.
  - Refactored legacy C++ code, improving modularity and enabling migration to new modernized architecture.
  - Optimized simulation workflows to cut performance failures and enhance overall reliability for on-premise and Cloud services.
  - Leading design of AI-powered productivity tools, projected to streamline debugging and optimize engineering time by 40%.
- **Fullstack Software Engineer Intern,** *Onset Data Loggers - LI-COR – Bourne, MA* (May 2024 - August 2024)
  - Built full-cycle MQTT handler processing > 10k IoT messages daily, accelerating real-time device-Cloud communication.
  - Developed UI tool for firmware team to upload, edit, and delete JSONs for the Cloud, reducing manual setup time by 60%.
  - Prototyped a customizable mapping dashboard using Leaflet and GIS APIs, enabling user interaction with live weather data.
- **Head Teaching Assistant,** *Brown University CS – Providence, RI* (October 2023 - May 2024)
  - Managed a 100-student upper-level CS course; recruited and led 14 TAs, standardizing grading and student support.
  - Redesigned course projects to align with real-world UI/UX workflows, improving student engagement and enthusiasm.
  - Coordinated 20+ industry professionals for portfolio review sessions, increasing student exposure to industry standards.
- **Frontend Software Engineer Intern,** *Onset Data Loggers – Bourne, MA* (May 2023 - August 2023)
  - Delivered 2 production-ready features ahead of schedule for an environment data collection and management platform.
  - Collaborated with middleware engineers to improve API efficiency and reduce UI latency.
  - Completed multiple user stories per sprint in Agile workflows, expediting iteration cycles and release cadence.
- **Founder & Content Creator,** *Independent Art Business* (2020 - 2023)
  - Built and scaled a digital art brand to 170k+ followers, reaching millions of views through data-driven content experimentation.
  - Produced and shipped physical products end-to-end, fulfilling 60+ orders and selling 300+ units for $1.6k profit within months.
  - Partnered with Huion, XPPen, and Artisul in official collaborations to create multi-platform marketing and advertising content.

RECENT PROJECTS:

- **Ninja Mouse (Custom 3D Game Engine)**
  - Developed a C++ 3D game engine with component-based architecture, triangle-based collision detection, and AI pathfinding.
  - Abstracted engine components into 17 modules with OOP principles to allow scalability and robust per-game customization.
  - Integrated custom 3D assets with material loading support and environment shaders for enhanced visual fidelity.
  - Leveraged BVH and grid-based acceleration trees to logarithmically optimize environment creation and collision detection.
- **3D Modeling & Animation**
  - Animated and imported game-ready and cyclable attack sequences with a custom 3D rig and character into Unreal.
  - Designed, modeled, rigged, textured, and animated 2 original characters with custom shaders for an animated short.
  - Created adaptable shader and texture customization sliders, supporting rapid creation of varying environments.
  - Led storyboarding, character design, and character modeling for a 3 minute animated short produced within 1 month.
- **Origin Two (Custom 2D Game Engine)**
  - Developed Java-based 2D game engine with collision detection, AI pathfinding, animations, and dynamic level loading.
  - Implemented a precise, boundary-based physics system and custom UI framework for seamless player interaction.
  - Scaled engine to support 3+ dungeon crawler and platformer games alongside integrated sound and text generation.
- **Realtime Ocean Water Simulation**
  - Implemented a physically accurate ocean water simulation based on Jerry Tessendorf's paper on motion-picture oceans.
  - Built a fast fourier transform optimizer to allow real-time user manipulation based on wind speed and turbulence meters.
  - Introduced new features ocean foam, particle systems for spray, caustics, and custom shaders for reflection and refraction.
- **Realtime Graphics & Ray Tracing**
  - Created a real-time C++/OpenGL scene viewer with shape and camera manipulation and custom GLSL shaders.
  - Integrated ray tracing algorithm for realistic rendering using the Phong model, mirror reflections, and texture mapping.
  - Programmed custom shaders with framebuffer objects to achieve stylized post-processing effects.

SKILLS:

- **Computer Languages:** C++, Python, Java, React.js, Vue.js, Node.js, JS, TS, Shell, HTML/CSS, OpenGL, TensorFlow, Scipy
- **Tools:** Maya, Blender, Adobe, Git, Docker, Figma, Microsoft Office (Word, PowerPoint, Excel), Kafka, NoSQL
- **Skills:** Atlassian/Jira, Agile, CI/CD, Scrum, Kanban, Project Management